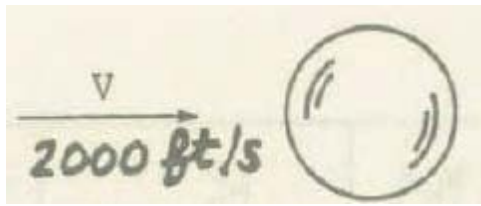
Problem 9.15

(Difficulty 1)

9.15 A steel sphere of 0.25 in diameter has a velocity of 200 $\frac{ft}{s}$ at an altitude of 30000 ft in the U.S. Standard Atmosphere. Calculate the drag force on this sphere.



Find: The drag force on the sphere

Assumption: The sphere velocity changes slowly enough so that the flow can be considered steady

Solution: Use the relations for drag coefficient

For the sphere, the drag coefficient is defined as

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2 A}$$

Where the area is the frontal area of the sphere. The drag coefficient is a function of Reynolds number, as shown in Figure 9.11. At 30000 ft attitude we have:

$$\rho = 0.000891 \frac{slug}{ft^3}$$

$$\mu = 3.107 \times 10^{-7} \frac{lbf \cdot s}{ft^2}$$

The Reynolds number is:

$$Re = \frac{\rho U D}{\mu} = \frac{0.000891 \frac{lbf \cdot s^2}{ft^4} \times 200 \frac{ft}{s} \times 0.0208\, ft}{3.107 \times 10^{-7} \frac{lbf \cdot s}{ft^2}} = 1.193 \times 10^4$$

From the figure 9.11 we have:

$$C_D = 0.4$$

The drag force is calculated as:

$$F_D = C_D \frac{\rho}{2} U^2 A = 0.4 \times \frac{1}{2} \times 0.000891 \ \frac{slug}{ft^3} \times \left(200 \ \frac{ft}{s}\right)^2 \times \frac{\pi}{4} \times (0.0208 \ ft)^2 = 0.00242 \ lbf$$

Problem 9.16

(Difficulty: 2)

9.16 Small oil droplets with a specific gravity of 85 rise in a 30 °C water bath. Determine the terminal speed of a droplet as a function of droplet diameter assuming the drag coefficient is given by the relation for Stokes flow. Determine the maximum droplet diameter for which Stokes flow is a reasonable assumption.

Assumptions: Stokes drag law is valid

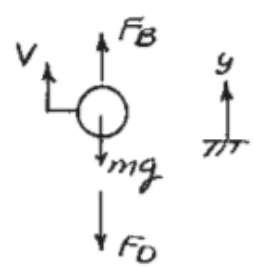
Solution: Apply Newton's second law and the definition of drag coefficient.

$$\sum F_y = -mg + F_B - F_D = ma_y$$

For Stokes Law, for $Re < 10$

$$F_D = 3\pi\mu VD$$

Draw free-body diagram of droplet,



Then, at the terminal speed V_t.

$$-\rho\forall g + \rho_{H_2o}\forall g - 3\pi\mu V_t D = 0$$

Solving, we have:

$$V_t = \frac{(\rho_{H_2o} - \rho_o)\forall g}{3\pi\mu D} = \rho_{H_2o}(1 - SG_o)\frac{\pi D^3}{6}\frac{g}{3\pi\mu D} = \frac{(1 - SG_o)D^2 g}{18\upsilon}$$

Evaluating,

$$V_t\ (m/s) = \frac{(1 - 0.85)}{18} \times (D\ mm)^2 \times 9.81\ \frac{m}{s^2} \times \frac{s}{1 \times 10^{-6}\ m^2} \times \frac{m^2}{10^6\ mm^2}$$

$$V_t\ (m/s) = 0.0818[D\ (mm)]^2$$

For Stokes flow, $Re < 1$, so

$$Re = \frac{\rho V_t D}{\mu} = \frac{V_t D}{\upsilon} = \frac{(1 - SG_o)D^3 g}{18\upsilon^2} \leq 1$$

Thus

$$D^3 \leq \frac{18\upsilon^2}{(1 - SG_o)g}$$

$$D \leq \left[\frac{18\upsilon^2}{(1 - SG_o)g}\right]^{\frac{1}{3}}$$

Evaluating,

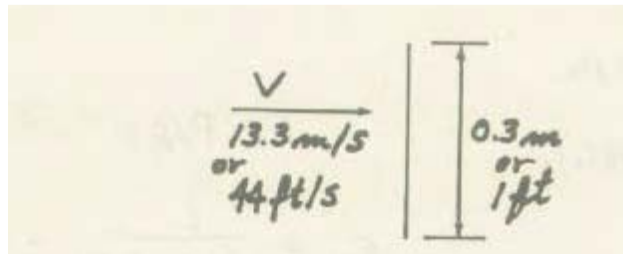
$$D \leq \left[\frac{18}{(1 - 0.85)} \times (1.00 \times 10^{-6})^2 \, \frac{m^4}{s^2} \times \frac{s^2}{9.81\, m}\right]^{\frac{1}{3}} = 2.31 \times 10^{-4}\, m \; (0.231\, mm)$$

Thus Stokes flow will be a valid assumption for $D < 0.231\, mm$.

Problem 9.20

(Difficulty 1)

9.20 A 0.3 m diameter circular disk is placed normal to a flow stream that has a speed of 5 m/s. Calculate the force and power if the fluid is (a) air and (b) water. If the disk is replaced with a hemisphere with the open end facing downstream, determine the power for the two fluids. Explain why there is a difference for the two geometries even though the frontal area is the same.



Find: The force and power for air and water

Solution: Use the drag coefficient to find the force and power

The drag coefficient is:

$$C_D = 1.12$$

The drag force is:

$$F_D = C_D \times \frac{1}{2}\rho A V^2$$

The power is:

$$P = F_D V$$

The velocity is:

$$V = 48\ \frac{km}{h} = 13.33\ \frac{m}{s}$$

The area is:

$$A = \frac{\pi}{4}D^2 = \frac{\pi}{4} \times (0.3\ m)^2 = 0.0707\ m^2$$

(a) For standard air at sea level:

$$\rho = 1.225\ \frac{kg}{m^3}$$

$$F_D = 1.12 \times \frac{1}{2} \times 1.225\ \frac{kg}{m^3} \times 0.0707\ m^2 \times \left(13.33\ \frac{m}{s}\right)^2 = 8.62\ N$$

$$P = F_D V = 8.62\ N \times 13.33\ \frac{m}{s} = 114.9\ W$$

(b) For water:

$$\rho = 998\ \frac{kg}{m^3}$$

$$F_D = 1.12 \times \frac{1}{2} \times 998\ \frac{kg}{m^3} \times 0.0707\ m^2 \times \left(13.33\ \frac{m}{s}\right)^2 = 7021\ N$$

$$P = F_D V = 7021\ N \times 13.33\ \frac{m}{s} = 93.6\ kW$$

Problem 9.22

[Difficulty: 2]

9.22 A projectile reduces its speed from 250 m/s to 210 m/s over a horizontal distance of 150 m. The diameter and mass of the projectile are 11 mm and 16 g, respectively. Determine the average drag coefficient for the bullet

Given: Ballistic data for .44 magnum revolver bullet

$$V_i = 250 \cdot \frac{m}{s} \quad V_f = 210 \cdot \frac{m}{s} \quad \Delta x = 150 \cdot m \quad M = 15.6 \cdot gm \quad D = 11.2 \cdot mm$$

Find: Average drag coefficient

Solution:

Basic equations: $F_D = C_D \cdot \frac{1}{2} \cdot \rho \cdot V^2 \cdot A$ (Drag)

Assumptions: (1) Standard air
(2) Neglect all losses other than aerodynamic drag

Newton's 2nd law: $\Sigma F_x = -F_D = M \cdot \left(\frac{d}{dt} V\right) = M \cdot V \cdot \left(\frac{d}{ds} V\right)$ Therefore: $M \cdot V \cdot \left(\frac{d}{dx} V\right) = -C_D \cdot \frac{1}{2} \cdot \rho \cdot V^2 \cdot A$

Separating variables: $dx = -\frac{2 \cdot M}{C_D \cdot \rho \cdot A} \cdot \frac{dV}{V}$ Integrating both sides yields: $\Delta x = -\frac{2 \cdot M}{C_D \cdot \rho \cdot A} \cdot \ln\left(\frac{V_f}{V_i}\right)$

Solving this expression for the drag coefficient: $C_D = -\frac{2 \cdot M}{\Delta x \cdot \rho \cdot A} \cdot \ln\left(\frac{V_f}{V_i}\right)$ The area is: $A = \frac{\pi}{4} \cdot D^2 = 98.52 \cdot mm^2$

Therefore the drag coefficent is: $C_D = 0.299$